

Zi Xuan Lim (she/her)

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RESEARCH PROFILE

High-achieving Computer Science researcher specializing in **Deep Learning**, **Computer Vision** and **Multimodal Vision-Language Models (VLMs)** for Intelligent Transportation Systems (ITS). Proven track record in developing localized open-source datasets and optimizing YOLO-based architectures for real-time deployment. Currently advancing research in automated traffic analysis and UAV-integrated disaster response, with manuscripts under review in Scopus-indexed journals. Seeking to leverage technical expertise in machine learning and efficiency-driven AI to pioneer innovative, scalable solutions through a Direct PhD in Computing at Sunway University.

ACADEMIC ACHIEVEMENTS

Sunway University

Bachelor of Science (Honours) in Computer Science
CGPA: 3.98/4.00

Bandar Sunway, Selangor

April 2023 - April 2026

Sunway College Kuala Lumpur

Foundation in Science and Technology (FIST)
CGPA: 4.00/4.00

Bandar Sunway, Selangor

April 2022 - March 2023

Sekolah Menengah Kebangsaan Yong Peng

Sijil Pelajaran Malaysia (SPM)
Grades: 10As (5A+, 2A, 3A-)

Batu Pahat, Johor

January 2017 - March 2022

HONOURS & AWARDS

Tan Sri Sir Jeffrey Cheah Scholastic Award & Lancaster University Chancellor's Medal

Awarded July 2026

Awarded for achieving the highest cumulative CGPA, accompanied by an RM2,000 cash prize:

- Scholastic Award: Awarded to the top graduating student in the BSc (Hons) in Computer Science programme.
- Chancellor's Medal: Awarded to the top graduating student across the Faculty of Engineering and Technology.

Jeffrey Cheah Foundation (JCF) Scholarship

Awarded November 2023

Awarded a full tuition scholarship valued at approximately RM100,000 based on outstanding academic achievement

Best Presentation Award - Women Research Engineers Network (WREN) Symposium

Awarded May 2025

Recognized for the research presentation "Advancing Smart Traffic Infrastructure with AI-based Vehicle Detection Across Computing Configurations". The award included an 1-on-1 academic career consultation with Dr. Shelda Debowski.

Sunway University President's Award

Awarded March 2023

Awarded to the highest-achieving Foundation in Science and Technology (FIST) student. Accompanied by an RM1,000 cash prize.

SPM Top Achiever Award

Awarded March 2022

Awarded by SMK Yong Peng for outstanding academic performance in the SPM 2021 examination.

PUBLICATIONS & RESEARCH PRESENTATIONS

Journal Manuscripts

- Teoh, J. **Lim, Z. X. (Co-first authors)**, Chan, Y. H., Kabir, A. S., Yap, K. M., "Recent Advancements, Challenges, and Future Directions of Integrating Drones in Smart City Traffic Management: A Systematic Review, Journal of Electrical and Computer Engineering, 2026, 7869329, 26 pages, 2026. <https://doi.org/10.1155/jece/7869329>.
- Teoh, J., **Lim, Z. X.**, Zulkoffli, Z. B., Chua, H. S., Lee, Y. L., and Yap, K. M., "A Systematic Review on Flood Response in Malaysia: Technical Advancements, Challenges and Opportunities." targeting *IEEE Access*. **(In Preparation)**

Conference Proceedings & Presentations

- **Lim, Z. X.**, Teoh, J. and Yap K. M., "Advancing Smart Traffic Infrastructure with AI-based Vehicle Detection Across Computing Configurations." *Women Research Engineers Network (WREN) Symposium 2025*, University of Wollongong (UoW), Australia.

- **Lim, Z. X.,** Teoh, J. and Yap K. M., “MY-VID: An Open-Source Malaysian Vehicle Image Dataset for Intelligent Transportation System and Road Safety.” *International Workshop on Advanced Image Technology (IWAIT) 2026. (Extended Abstract Accepted)*

Open-Source Datasets

- **Lim, Z. X.,** Teoh, J. and Yap K. M., “MY-VID: An Open-Source Malaysian Vehicle Image Dataset for Intelligent Transportation System and Road Safety.” *Zenodo* (2025). <https://doi.org/10.5281/zenodo.16866508>.

PROFESSIONAL EXPERIENCE

GoLearn Education Sdn. Bhd.

Computer Science Tutor (Part-Time)

June 2026 - Present

- **Curriculum & Instruction:** Teach CS fundamentals based on Malaysian KSSM and Cambridge IGCSE syllabi, simplifying complex algorithms and flowcharts for secondary students
- **Personalized Mentorship:** Design customized lesson plans and practical exercises to continuously track and improve individual student progress

Independent Full-Stack Developer

May 2026 - July 2026

Freelance Web Development

- **Frontend Architecture & UI/UX:** Designed and developed a responsive, highly optimized booking website for a luxury homestay utilizing a modern tech stack of React, TypeScript, and Vite
- **Backend Integration & Automation:** Integrated seamless booking inquiry forms paired with serverless email automation to streamline customer communications and reservation workflows
- **Localization & Edge Optimization:** Implemented full multilingual support and optimized the cross-device user experience, ensuring high performance and accessibility for international mobile and desktop users

WISE AI Malaysia Sdn. Bhd.

AI Engineer Intern

January 2026 – April 2026

- **Face Quality Classification:** Fine-tuned a face quality and eye-closure classifier using MobileNet and MediaPipe, benchmarking against Qwen-VL and YOLO architectures
- **Interactive Deployment:** Deployed the eye-state classification model using Gradio, creating an interactive, real-time visualization interface to effectively monitor and validate user eye-closure behaviors
- **MLOps Infrastructure:** Built a FastAPI backend for automated data ingestion and preprocessing, managing GPU allocation and containerization via Docker
- **Edge Optimization:** Applied FP16 and INT8 Post-Training Quantization (PTQ) for ONNX and OpenVINO deployment, boosting inference speed by 40% while evaluating any accuracy loss exceeding -1.5% as severe
- **Mentorship:** Supervised an intern on data annotation pipelines for eKYC, face liveness, and ID fraud detection systems

Research Centre for Human-Machine Collaboration (HUMAC)

UG Lab Assistant

June 2024 – January 2026

- **Academic Support:** Assisted PhD candidate in research project integrating computer vision into UAVs for Malaysian flood rescue operations
- **UAV Integration:** Designed and assembled a custom-built drone, including hardware setup, circuit integration, and flight software configuration to support data collection for research
- **Algorithm Benchmarking:** Conducted comparative research on object detection architectures, including YOLO, Faster R-CNN, and Detection Transformers (DETR), to determine optimal accuracy-latency trade-offs for real-time UAV applications
- **Research Leadership & Publications:** Co-led brainstorming, ideation, and task delegation for smart city traffic management research, resulting in a co-first authored publication in a **Q2 journal** (Journal of Electrical and Computer Engineering). Currently spearheading a second systematic review on flood response technologies targeting a **Q1 journal** (IEEE Access).

TECHNICAL SKILLS

- **Deep Learning Frameworks:** PyTorch, TensorFlow, Scikit-learn, YOLO, Faster R-CNN, DETR, MobileNet
- **Computer Vision Tools:** OpenCV, MediaPipe, Roboflow, LabelMe, Image Augmentation, Model Evaluation
- **Multimodal & Generative AI:** Vision-Language Models (VLMs), Qwen-VL, Qwen-Image-Edit
- **Deployment & DevOps:** FastAPI, Gradio, Streamlit, Quantization, Docker, Conda, Git/GitHub, Linux/Ubuntu
- **Programming & Databases:** Python, Java, Scala, SQL (Oracle/MySQL), MongoDB
- **Hardware:** UAV Assembly, Drone Circuit Integration